Epidemiology

Event Category: Emergency Preparedness

Eligible Divisions: Secondary & Postsecondary/ Collegiate	**Round One:** 50 Q test (60 minutes)	**Digital Upload:** NO
Solo Event: 1 competitor		

New for 2026-2027

Editorial changes have been made.

Event Summary

Epidemiology allows members to gain knowledge about health and disease in populations. This competitive event consists of a written exam of concepts related to the study of epidemiology. In the event of a tie, the essay will be judged to break the tie. Competitors must recognize, identify, define, interpret, and apply these concepts in a multiple-choice test. The event aims to motivate members to become proactive future health professionals, enhancing their scientific literacy and offering valuable insights into careers in public health.

Sponsorship

HOSA-Future Health Professionals appreciates the sponsorship of this event by ASPPH.	ASPPH ASSOCIATION OF SCHOOLS & PROGRAMS OF PUBLIC HEALTH

General Event Information

General Rules	Competitors must be familiar with and adhere to the General Rules and Regulations.
Dress Code	Proper business attire or official HOSA uniform. Bonus points will be awarded for proper dress.
Competitors Must Provide	● Photo ID ● Two #2 lead pencils (not mechanical) with an eraser ● Neon colored foam earplugs, NOT earbuds, for noise control (optional) HOSA Conference Staff will provide equipment and supplies listed in Appendix I.
Official References	The references below are used in the development of the test questions. Content is posted as of September 1, 2026: ● PH 101 Series ● ActivEpi ● Environmental Medicine Education Resources ● CDC: Morbidity and Mortality Weekly Report

ROUND ONE- Test

Test Instructions	All competitors will be given a test and test answer sheet. The written test consists of 50 multiple-choice items and is administered within a 60-minute time frame. See test instructions here. One essay question will be administered with the original test.

The expectation is that competitors read and are aware of all the content within these guidelines and associated links. Successful competitors will study all links for the detailed information.

	At the International Leadership Conference, HOSA will provide basic handheld calculators (no graphing calculators) for addition, subtraction, division, multiplication, and square root calculations. Personal calculators are NOT allowed.
Time Remaining Announcements	There will be NO verbal announcements for time remaining during ILC testing. All ILC testing will be completed in the Testing Center, and competitors are responsible for monitoring their own time.
Test Plan	The test plan for the Epidemiology Test is: ● Introduction to Epidemiology: (30%) ○ Basic concept ○ Types of case studies ○ Measuring health and disease ○ Causation ○ Surveillance ● World and Government Response to Infectious Diseases (20%) ○ WHO Infectious Diseases ○ CDC Activities ○ Government Responses ● CDC: Morbidity and Mortality Weekly Report (50%) ○ Case studies, investigations, surveillance summaries, scientific information recommendations
Sample Test Questions	1. What is the term for the proportion of a disease or other outcome in exposed individuals that can be credited to the exposure? (WHO pp 35) **A. Attributable risk** B. Risk benefit C. Cohort study D. Odds advantage 2. In an epidemiology study, how is the specified population identified? (CDC Intro 1-4) A. People affected by the condition/disease B. Conditions where large numbers of people were affected **C. People exposed to a condition/disease** D. People who died from a disease/condition 3. Which type of epidemiological study is best characterized as a survey? (CDC ATSDR) A. Case-control B. Cohort C. Relational **D. Cross-sectional**

Final Scoring

In the event of a tie, the essay will be judged to break the tie.

Future Opportunities

The HOSA Scholarship program has funding available for HOSA members who have an interest in various health careers! The application process begins January 1, 2027, and can be found here: https://hosa.org/scholarships/.

Graduating from high school or completing your postsecondary/collegiate program does not mean your HOSA journey has to end. As a HOSA member, you are eligible to become a HOSA Lifetime Alumni Member- a free and valuable opportunity to remain connected, give back, and help to shape the future of the organization. Learn more and sign up at hosa.org/alumni.